Immediately before the first pulse of light reaches the floor, drop the floor clock into the hole (diagram b). The second pulse arrives so soon afterward that the freely falling floor clock has moved imperceptibly between pulses, and is still very nearly at rest with respect to the floor, and therefore is still feeling the same flow of time as does the floor itself.

In this way, Einstein converted the problem of comparing the flow of time as felt by the ceiling and the floor into the problem of comparing the ticking rates of two freely falling clocks: the falling ceiling clock which feels ceiling time, and the falling floor clock which feels floor time. Einstein's equivalence principle then permitted him to compare the ticks of the freely falling clocks with the aid of his special relativistic laws.

Because the ceiling clock was dropped before the floor clock, its downward speed is always greater than that of the floor clock (diagram b); that is, it moves toward the floor clock. This implies that the floor clock will see the ceiling clock's light pulses *Doppler-shifted* (Box 2.3); that is, it will see them arrive more closely spaced in time than the time between its own ticks. Since the time between pulses was regulated by the ceiling's time flow, and the time between floor-clock ticks is regulated by the floor's time flow, this means that time must flow more slowly near the floor than near the ceiling; in other words, *gravity must dilate the flow of time.*

On December 24, he wrote to a friend saying, "At this time I am busy with considerations on relativity theory in connection with the law of gravitation ... I hope to clear up the so-far unexplained secular changes of the perihelion shift of Mercury . . . but thus far it does not seem to work." By early 1908, frustrated by no real progress, Einstein gave up, and turned his attention to the realm of atoms, molecules, and radiation (the "realm of the small"), where the unsolved problems for the moment seemed more tractable and interesting.[3]

Through 1908 (while Minkowski unified space and time, and Einstein pooh-poohed the unification), and through 1909, 1910, and 1911, Einstein stayed with the realm of the small. These years also saw him move from the patent office in Bern to an associate professorship at the University of Zurich, and a full professorship in Prague—a center of the Austro-Hungarian empire's cultural life.